YuJia Liang

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Education

National Taiwan Ocean University — Keelung, Taiwan

B.S. in Computer Science and Engineering — *Sep. 2022 – Jun. 2026*

Graduate-Level Coursework (9 courses, 27 credits): Machine Learning, Generative AI, Natural Language Processing, Data Mining, Information Retrieval, Advanced Database Systems, Formal Methods, Smart IoT Platform

Relevant Courses: Data Structures & Algorithms, Graph Theory Algorithms, Operating Systems, Computer Organization, Computer Networks, Verilog HDL, Digital Logic, Linear Algebra, Probability Theory, Discrete Mathematics

Awards & Honors

Apple WWDC26 Swift Student Challenge — Winner 2026

One of 8 winners in Taiwan and 350 worldwide; covered by CNA, UDN, Liberty Times, ETtoday, and TechNews

Silver Award, Department of CSE Software Development Award 2026

Volleyball Performance Analysis System — led a 3-person team — *Keelung, Taiwan*

Experience

ITRI (Industrial Technology Research Institute) — Jun. 2025 – Aug. 2025

AI Image Recognition Intern, Electronics and Optoelectronics Research Laboratories — *Hsinchu, Taiwan*

- Trained **YOLOPX**, a multi-task panoptic driving perception network performing object detection, drivable-area segmentation, and lane segmentation in a single forward pass; reached **0.972 mIoU** (drivable area), **0.793 mIoU** (lane), and **0.838 mAP@0.5** over 4 classes after 200 epochs on ITRI's in-house driving dataset
- Extended the open-source YOLOPX codebase by **∼1,900 lines** (6 files rewritten, 5 added) for ITRI's data format, building a YOLO-segmentation loader and a Mosaic/MixUp/Albumentations augmentation pipeline to replace one that handled BDD100K only
- Owned the **edge deployment path**: applied network pruning, exported to ONNX, wrote a tool resolving dynamic convolution-bias subgraphs that blocked TensorRT conversion, built a **TensorRT/PyCUDA engine for NVIDIA Jetson Nano**, and streamed images and detections from it to a desktop PC over **Ethernet/ROS at 30 FPS with sub-50ms latency**
- Delivered a command-line **batch-inference tool for ARTC** (Automotive Research & Testing Center) running the trained model across image and video datasets including a new rainy-weather set; separately reproduced and benchmarked **YOLOP, YOLOPv2, YOLOPX, and A-YOLOM** to identify deployment-viable designs, presenting findings at weekly research reviews

Research

ToneLeak — Counterfactual Auditing of Guitar Tone Models 2026

Python, PyTorch, scikit-learn, librosa, NumPy — *Independent Research*

- Testing whether guitar-tone models alter performance attributes that should stay attributable to the player (bends, vibrato, attack, dynamics), via a **paired counterfactual design** where a deterministic effect on the identical DI performance supplies ground truth
- Audited **2,040 published 64-dim Fx embeddings** over 680 renders of the public Open-Amp encoder: tone decodes at **99.3% balanced accuracy** but source combination stays at **87.3%** — tone-informative yet **not source-invariant**; a held-out causal reference-swap confirmed the output shifts under an identical query and tone target (log-spectral RMSE 0.538, 95% CI [0.461, 0.627])
- Reported a **negative result**: projecting out 5 source-predictive dimensions cut source decodability from 82.6% to 47.3% but raised same-DI target error 51–82%, so it is not a viable mitigation; grounded the claim in a documented novelty audit against ISMIR/DAFx/ICASSP prior work

Projects

Volleyball Performance Analysis System – Senior Capstone — Nov. 2024 – Dec. 2025

Python, PyTorch, YOLO, U-Net/ONNX, OpenCV, FastAPI, Celery, Docker — *Advisor: Prof. Pei-Yi Ding*

- Developed comprehensive volleyball analytics system achieving 94.49% mAP@0.5 for action recognition (5 classes: serve, spike, block, receive, set) and 79.5% ball tracking accuracy

- Architected distributed microservices with FastAPI backend, Celery task queue, and Redis message broker; implemented WebSocket endpoints for real-time analysis progress tracking
- Curated a 24,806-image training set by merging public volleyball datasets with ∼12,000 frames annotated in-house, training the action classifier to 94.46% precision / 91.52% recall
- Built a multi-model pipeline: VballNet (U-Net ONNX) for ball tracking at 200+ FPS, YOLOv11m for action detection, YOLOv8+Norfair for player tracking at 87.6% ID consistency, with trajectory smoothing and an action-consolidation algorithm cutting fragmented events by 45%
- Containerized 5 services with Docker Compose and GitHub Actions CI/CD, a pytest suite over API, database and ML inference paths, and a React/TypeScript dashboard with frame-level timeline seeking

RiffNode – AI Guitar Effects Studio for iPadOS & macOS 2026

Swift 6, SwiftUI, AVFoundation, Accelerate/vDSP, Foundation Models *WWDC26 Swift Student Challenge*

- Built a guitar effects studio entirely from native Apple frameworks, organised into a **Clean Architecture** split (Domain/Application/Infrastructure/Presentation) with dependencies pointing inward — **WWDC26 Swift Student Challenge Winner**
- Engineered a low-latency `AVAudioEngine` pipeline driving 11 effects in a drag-and-drop chain under **Swift 6 strict concurrency**; keeping every pedal wired to the graph and treating enabled state as a bypass flag makes toggles glitch-free and confines rebuilds to reordering, while Accelerate/vDSP FFT and autocorrelation pitch detection drive a tuner, metronome, and chord detection
- Rebuilt the assistant as a **tool-calling agent** over 13 rig tools with a multi-turn observe-act-report loop and undo, clamping every model-proposed value to its `ParameterSpec` range so no model output reaches the DSP unchecked; one inference protocol backs **5 interchangeable providers** (on-device Apple Foundation Models by default and fully offline, Claude Code CLI, Codex CLI, Anthropic API, keyword fallback) with retrieval over an on-device NaturalLanguage embedding corpus

Soredemo – Release-to-Demo Automation for macOS 2026

Swift 6, SwiftUI, ScreenCaptureKit, AVFoundation, Core Image, SwiftPM plugins *Individual Project*

- Architected an **11-target Swift package in three tiers**: a deterministic core (same input, same output; no GPU, clock or window), a Capture tier over ScreenCaptureKit and accessibility APIs, and a GPU Compositing tier — with the latter two forbidden from depending on each other
- Validated the product premise empirically rather than assuming it, measuring **2,241 GitHub releases across 151 developer tools** and finding only 2.1% carried any visual demo
- Wrote **two custom SwiftPM build plugins** turning those boundaries into compile errors rather than review conventions, held **Swift 6 strict memory safety** with zero unmarked unsafe sites, and wrote **584 tests with more test code than source** (10,849 vs 10,171 lines), partitioned so a pure-tier test cannot link a target that reads the world

Technical Skills

Programming: Python, C++, Rust, Swift, TypeScript/JavaScript, Dart, SQL
ML & Computer Vision: PyTorch, TensorFlow, OpenCV, ONNX Runtime, TensorRT, ROS, Jetson Nano, model pruning & quantization, RAG & embedding retrieval, scikit-learn, NumPy, librosa
Apple Platforms: SwiftUI, AVFoundation, Accelerate/vDSP, ScreenCaptureKit, Core Image, NaturalLanguage, Foundation Models
Web & Backend: React, Next.js, Tailwind CSS, TypeScript, FastAPI, Celery, Flutter
Data & Infrastructure: Docker, GitHub Actions, GCP, Git, Redis, PostgreSQL, MongoDB, Firebase, pytest, SwiftPM
Languages: Mandarin (Native), English (Fluent, TOEFL 110, TOEIC 920), Japanese (JLPT N3)

Leadership & Activities

Volleyball Department Team Captain 2022 – present

Led the team to the top 8 at the national Information Departments Cup *Keelung, Taiwan*

NTUFC Kronos Quant Club — Team Pyxis Sep. – Dec. 2025

3 CTA alphas on OKX perpetuals — 283% in a 22-month backtest (Sharpe 4.76), deployed on GCP

NTU Photography Club (NTUPhoto) 2025 – present

Member & Darkroom Officer *Taipei, Taiwan*

Hackathon Participant 2022 – present

Hotai Motor and TSMC corporate hackathons